

Mini_Project3

September 23, 2026

```
[1]: pip install pandas
```

```
Requirement already satisfied: pandas in C:\Users\surfce\anaconda3\Lib\site-  
packages (3.0.3)  
Requirement already satisfied: numpy>=2.3.3 in  
C:\Users\surfce\anaconda3\Lib\site-packages (from pandas) (2.4.6)  
Requirement already satisfied: python-dateutil>=2.8.2 in  
C:\Users\surfce\anaconda3\Lib\site-packages (from pandas) (2.9.0.post0)  
Requirement already satisfied: tzdata in C:\Users\surfce\anaconda3\Lib\site-  
packages (from pandas) (2026.2)  
Requirement already satisfied: six>=1.5 in C:\Users\surfce\anaconda3\Lib\site-  
packages (from python-dateutil>=2.8.2->pandas) (1.17.0)  
Note: you may need to restart the kernel to use updated packages.
```

```
[2]: pip install numpy
```

```
Requirement already satisfied: numpy in C:\Users\surfce\anaconda3\Lib\site-  
packages (2.4.6)  
Note: you may need to restart the kernel to use updated packages.
```

```
[3]: pip install matplotlib scikit-learn
```

```
Requirement already satisfied: matplotlib in C:\Users\surfce\anaconda3\Lib\site-  
packages (3.11.0)  
Requirement already satisfied: scikit-learn in  
C:\Users\surfce\anaconda3\Lib\site-packages (1.9.0)  
Requirement already satisfied: contourpy>=1.0.1 in  
C:\Users\surfce\anaconda3\Lib\site-packages (from matplotlib) (1.3.3)  
Requirement already satisfied: cycycler>=0.10 in  
C:\Users\surfce\anaconda3\Lib\site-packages (from matplotlib) (0.12.1)  
Requirement already satisfied: fonttools>=4.22.0 in  
C:\Users\surfce\anaconda3\Lib\site-packages (from matplotlib) (4.63.0)  
Requirement already satisfied: kiwisolver>=1.3.1 in  
C:\Users\surfce\anaconda3\Lib\site-packages (from matplotlib) (1.4.9)  
Requirement already satisfied: numpy>=1.25 in  
C:\Users\surfce\anaconda3\Lib\site-packages (from matplotlib) (2.4.6)  
Requirement already satisfied: packaging>=20.0 in  
C:\Users\surfce\anaconda3\Lib\site-packages (from matplotlib) (26.0)  
Requirement already satisfied: pillow>=9 in C:\Users\surfce\anaconda3\Lib\site-
```

```

packages (from matplotlib) (12.3.0)
Requirement already satisfied: pyparsing>=3 in
C:\Users\surfce\anaconda3\Lib\site-packages (from matplotlib) (3.2.5)
Requirement already satisfied: python-dateutil>=2.7 in
C:\Users\surfce\anaconda3\Lib\site-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: scipy>=1.10.0 in
C:\Users\surfce\anaconda3\Lib\site-packages (from scikit-learn) (1.18.0)
Requirement already satisfied: joblib>=1.4.0 in
C:\Users\surfce\anaconda3\Lib\site-packages (from scikit-learn) (1.5.3)
Requirement already satisfied: narwhals>=2.0.1 in
C:\Users\surfce\anaconda3\Lib\site-packages (from scikit-learn) (2.23.0)
Requirement already satisfied: threadpoolctl>=3.5.0 in
C:\Users\surfce\anaconda3\Lib\site-packages (from scikit-learn) (3.5.0)
Requirement already satisfied: six>=1.5 in C:\Users\surfce\anaconda3\Lib\site-
packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
Note: you may need to restart the kernel to use updated packages.

```

```
[4]: pip install seaborn
```

```

Requirement already satisfied: seaborn in C:\Users\surfce\anaconda3\Lib\site-
packages (0.13.2)
Requirement already satisfied: numpy!=1.24.0,>=1.20 in
C:\Users\surfce\anaconda3\Lib\site-packages (from seaborn) (2.4.6)
Requirement already satisfied: pandas>=1.2 in
C:\Users\surfce\anaconda3\Lib\site-packages (from seaborn) (3.0.3)
Requirement already satisfied: matplotlib!=3.6.1,>=3.4 in
C:\Users\surfce\anaconda3\Lib\site-packages (from seaborn) (3.11.0)
Requirement already satisfied: contourpy>=1.0.1 in
C:\Users\surfce\anaconda3\Lib\site-packages (from
matplotlib!=3.6.1,>=3.4->seaborn) (1.3.3)
Requirement already satisfied: cycler>=0.10 in
C:\Users\surfce\anaconda3\Lib\site-packages (from
matplotlib!=3.6.1,>=3.4->seaborn) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in
C:\Users\surfce\anaconda3\Lib\site-packages (from
matplotlib!=3.6.1,>=3.4->seaborn) (4.63.0)
Requirement already satisfied: kiwisolver>=1.3.1 in
C:\Users\surfce\anaconda3\Lib\site-packages (from
matplotlib!=3.6.1,>=3.4->seaborn) (1.4.9)
Requirement already satisfied: packaging>=20.0 in
C:\Users\surfce\anaconda3\Lib\site-packages (from
matplotlib!=3.6.1,>=3.4->seaborn) (26.0)
Requirement already satisfied: pillow>=9 in C:\Users\surfce\anaconda3\Lib\site-
packages (from matplotlib!=3.6.1,>=3.4->seaborn) (12.3.0)
Requirement already satisfied: pyparsing>=3 in
C:\Users\surfce\anaconda3\Lib\site-packages (from
matplotlib!=3.6.1,>=3.4->seaborn) (3.2.5)
Requirement already satisfied: python-dateutil>=2.7 in

```

C:\Users\surface\anaconda3\Lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (2.9.0.post0)
Requirement already satisfied: tzdata in C:\Users\surface\anaconda3\Lib\site-packages (from pandas>=1.2->seaborn) (2026.2)
Requirement already satisfied: six>=1.5 in C:\Users\surface\anaconda3\Lib\site-packages (from python-dateutil>=2.7->matplotlib!=3.6.1,>=3.4->seaborn) (1.17.0)
Note: you may need to restart the kernel to use updated packages.

```
[5]: import pandas as pd
```

```
[6]: data=pd.read_csv(r"C:\Users\surface\Downloads\trends.csv")
```

```
[7]: data.head()
```

```
[7]:
```

	location	year	category	rank	query
0	Global	2001	Consumer Brands	1	Nokia
1	Global	2001	Consumer Brands	2	Sony
2	Global	2001	Consumer Brands	3	BMW
3	Global	2001	Consumer Brands	4	Palm
4	Global	2001	Consumer Brands	5	Adobe

```
[8]: data.tail()
```

```
[8]:
```

	location	year	category	rank	query
26950	Vietnam	2020	Là Gì?	1	Virus Corona là gì
26951	Vietnam	2020	Là Gì?	2	Miền thị thực là gì
26952	Vietnam	2020	Là Gì?	3	Đầu cắt moi là gì
26953	Vietnam	2020	Là Gì?	4	Bệnh bạch hầu là gì
26954	Vietnam	2020	Là Gì?	5	Đông Lào là gì

```
[9]: data.info
```

```
[9]: <bound method DataFrame.info of
```

	location	year	category	rank	query
0	Global	2001	Consumer Brands	1	Nokia
1	Global	2001	Consumer Brands	2	Sony
2	Global	2001	Consumer Brands	3	BMW
3	Global	2001	Consumer Brands	4	Palm
4	Global	2001	Consumer Brands	5	Adobe
...	...	...	...	...	...
26950	Vietnam	2020	Là Gì?	1	Virus Corona là gì
26951	Vietnam	2020	Là Gì?	2	Miền thị thực là gì
26952	Vietnam	2020	Là Gì?	3	Đầu cắt moi là gì
26953	Vietnam	2020	Là Gì?	4	Bệnh bạch hầu là gì
26954	Vietnam	2020	Là Gì?	5	Đông Lào là gì

```
[26955 rows x 5 columns]>
```

```
[10]: data.describe()
```

```
[10]:
```

	year	rank
count	26955.000000	26955.00000
mean	2015.243369	3.00000
std	3.564683	1.41424
min	2001.000000	1.00000
25%	2013.000000	2.00000
50%	2016.000000	3.00000
75%	2018.000000	4.00000
max	2020.000000	5.00000

```
[11]: data.shape
```

```
[11]: (26955, 5)
```

```
[12]: data.columns
```

```
[12]: Index(['location', 'year', 'category', 'rank', 'query'], dtype='str')
```

```
[13]: data.dtypes
```

```
[13]: location      str
year          int64
category       str
rank          int64
query         str
dtype: object
```

```
[14]: data.isnull().sum()
```

```
[14]: location      0
year            0
category        0
rank            0
query           0
dtype: int64
```

```
[15]: int(data.duplicated().sum())
```

```
[15]: 10
```

```
[16]: data.drop_duplicates(inplace=True)
```

```
[17]: data.nunique()
```

```
[17]: location      83
year           20
```

```
category      2450
rank           5
query        18431
dtype: int64
```

```
[18]: column_name='rank'
```

```
[19]: Q1=data['rank'].quantile(0.25)
```

```
[20]: Q3=data['rank'].quantile(0.75)
```

```
[21]: IQR=Q3-Q1
```

```
[22]: print(Q1)
```

```
2.0
```

```
[23]: print(Q3)
```

```
4.0
```

```
[24]: print(IQR)
```

```
2.0
```

```
[25]: LF=Q1-1.5*IQR
```

```
[26]: UF=Q3+1.5*IQR
```

```
[27]: print(LF)
```

```
-1.0
```

```
[28]: print(UF)
```

```
7.0
```

```
[29]: outliers=data[(data['rank']<LF) | (data['rank']>UF)]
```

```
[30]: print(f"Number of outliers: {len(outliers)}")
```

```
Number of outliers: 0
```

```
[31]: display(outliers)
```

```
Empty DataFrame
Columns: [location, year, category, rank, query]
Index: []
```

```
[32]: column_name1='year'
```

```
[33]: Q1=data['year'].quantile(0.25)
```

```
[35]: Q3=data['year'].quantile(0.75)
```

```
[36]: IQR=Q3-Q1
```

```
[37]: print(Q1)
      print(Q3)
      print(IQR)
```

```
2013.0
2018.0
5.0
```

```
[38]: LF=Q1-1.5*IQR
```

```
[39]: UF=Q3+1.5*IQR
```

```
[40]: print(LF)
```

```
2005.5
```

```
[41]: print(UF)
```

```
2025.5
```

```
[42]: outliers=data[(data['year']<LF) | (data['year']>UF)]
```

```
[43]: print(f"Number of outliers: {len(outliers)}")
```

```
Number of outliers: 480
```

```
[44]: display(outliers)
```

	location	year	category	rank	query
0	Global	2001	Consumer Brands	1	Nokia
1	Global	2001	Consumer Brands	2	Sony
2	Global	2001	Consumer Brands	3	BMW
3	Global	2001	Consumer Brands	4	Palm
4	Global	2001	Consumer Brands	5	Adobe
..	...	...	...	...	...
475	Global	2005	Most Searched	1	Janet Jackson
476	Global	2005	Most Searched	2	Hurricane Katrina
477	Global	2005	Most Searched	3	Tsunami
478	Global	2005	Most Searched	4	Xbox 360
479	Global	2005	Most Searched	5	Brad Pitt

```
[480 rows x 5 columns]
```

```
[ ]:
```